

Now calibration instruments and T&M devices come equipped with LAN and USB support. LAN and USB connectivity options are cheaper while offering equal or faster speed. There are solutions that allow only GPIB-compatible calibration devices to connect via USB, one example being Agilent's 82357A USB/GPIB interface. Agilent's E5810A LAN/GPIB interface allows remote control of a GPIB instrument through LAN. Many industry players are coming up with such solutions.

Automation software and remote calibration. Software suites are upgrading the level of automation in T&M calibration. Tripathi says, "Almost 95 per cent of the products are automated to reduce the measurement time and turn-around."

Not only test data generation and reporting but certain manual steps in test device calibration are also becoming automated. Nambiar cites an example: "National Instrument's Calibration Station leverages our in-house-developed Calibration Executive Software. It uses software routines to automate procedures and replace many manual calibration steps. The software also includes an advanced reporting feature. An example application is automatically recording the as-found and as-left states of each device. This means the same calibration team can accomplish many more calibrations in a given time period."



Fig. 2: Keysight's calibration kit



Fig. 3: National Instrument's GPIB/USB cable

Giving the example of Testo Industrial Services GmbH's move towards remote calibration, Bhangare says, "Testo offers a test equipment management system called PRIMAS. It is a combination of calibration and documentation management. PRIMAS online allows access to the calibration and test equipment data via the Internet. One can call up, modify or create data mobile via the Internet browser. Each calibration certificate is available to you by default as a PDF file."

Higher number of parallel device calibration. Touch-screen devices have found their way into calibration processes as well. Tools are coming with bigger capacities and volumes. Bhangare informs, "Conventional calibrators are improving. We now have huminators (humidity generators) with touch screens, data storage facility, USB connections and so on. Initially, calibrators had a small chamber space, but the latest huminators have sophisticated humidity chambers with compact design and extra space to calibrate more number of instruments at the same time."

Nambiar adds, "New technologies are making calibration tools more efficient and easy to use. Due to these improvements, the same teams are able to carry out even higher number of calibrations at the same time."

When should you calibrate your test device?

While general-purpose T&M devices should be calibrated at least once a

year, equipment for critical applications might require frequent servicing. Tripathi agrees, "Calibration requirement is directly proportional to the usage environment. Factors like workplace and usage frequency play a key role. It is advisable to calibrate devices at a regular interval of one year. If usage frequency is higher, the cycle can be reduced to ensure the quality." Device performance history will also give users an idea about the requirement of calibration.

Going forward

With a lot of R&D in this segment, industry players are attempting to further reduce the service time and enhance user experience. Predicting the future trends, Tripathi says, "Calibration devices will become more user-friendly. Touch screens on the equipment are providing easier access and control. Processes will become faster. Internal calibration methods will enable us to evaluate the equipment performance at an instance, covering just a few major parameters."

Connectivity technologies and standards are evolving, and so is the complexity of solutions. Nambiar quotes, "With the launch of new and complex technologies in all industries—be it 5G in telecommunications or advanced driver assistance systems in automotive—the sophistication of T&M instruments used is rising. As a result, calibration tasks for these instruments are becoming more complex. At the same time, there is demand for optimal results in a shorter period of time. Technologies like automation and cloud connectivity will continue to grow. With time, we will see more integration of automation software with the cloud from other companies as well. This will enable instantaneous monitoring of calibration reports, allowing technical teams to carry out analysis and preventative maintenance much faster." **EFY**